Course notes

Antiderivatives on an interval

We assume at this stage that every function continuous on an interval has antiderivatives; the link with area will be established in chapter 8. An antiderivative of f over an interval I is a differentiable function F such that $F' = f$ on I. If F is an antiderivative, all the antiderivatives on I are $F + C$, $C \in \mathbb{R}$.

Proof. If $F' = G' = f$, then $(F - G)' = 0$ on the interval; $F - G$ is therefore constant. Conversely, $(F + C)' = f$. The interval assumption is essential: on a disconnected domain, the constants may differ from one interval to another.

Table of antiderivatives - read the derivatives table backwards

$f(x)$	$F(x)$
$x^n \ (n \in \mathbb{Z},\ n \neq -1)$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x}$	$\ln \lvert x \rvert$ on an interval avoiding 0
$\frac{1}{\sqrt{x}}$	$2\sqrt{x} \quad (x > 0)$
e^x	e^x
$\cos x$	$\sin x$
$\sin x$	$-\cos x$

The additive constant can always be added. The last two lines will be reused after studying trigonometry.

Recognising a composition

For a function u differentiable, the antiderivatives of $u'u^n$, $u'e^u$, $\frac{u'}{u}$ are respectively $\frac{u^{n+1}}{n+1}$ $(n \neq -1)$, e^u, $\ln \lvert u \rvert$, on intervals where the expressions are defined. Linearity handles sums.

Example. An antiderivative of $6\,x(x^2+1)^2$ is $(x^2+1)^3$; an antiderivative of e^{3x} is $\frac{e^{3x}}{3}$. To impose $F(0) = 2$ in the latter case, take $F(x) = \frac{e^{3x}}{3} + \frac{5}{3}$.

Differential equations

A differential equation links an unknown function to its derivatives. A solution must be checked on a specified interval.

Standard solutions

$y' = ay$ has solutions $y(x) = Ce^{ax}$. For $a \neq 0$, $y' = ay + b$ has solutions

$$y(x) = Ce^{ax} - \frac{b}{a}.$$

If $a = 0$, the equation becomes $y' = b$ and $y(x) = bx + C$. If g is a known particular solution of $y' = ay + f(x)$, the solutions are $y = g + Ce^{ax}$.

Proof for $y' = ay$. If y is a solution, $z(x) = e^{-ax}y(x)$ satisfies $z' = e^{-ax}(y' - ay) = 0$; $z = C$. Conversely, the derivative of Ce^{ax} is aCe^{ax}. For $y' = ay + b$, subtract the constant solution $-\frac{b}{a}$ reduces to the homogeneous equation. For a variable right-hand side, do the same with g.

Example. $y' = -2y + 6$ and $y(0) = 1$ give $y = 3 + Ce^{-2x}$, then $C = -2$. The solution $3 - 2e^{-2x}$ increases to the equilibrium 3 when $x \to +\infty$.

Euler's method

For $y' = ay + b$, with a step $h > 0$, we approximate $y(t_k)$ by Y_k using $t_{k+1} = t_k + h$, $Y_{k+1} = Y_k + h(aY_k + b)$. This is an approximation using the tangent, not an equality for the exact solution.
A smaller step can improve the approximation; check the error numerically on a known solution.

Differentiating backwards and modelling change

7.1 Recognising and checking an antiderivative

First use: S52.

Definition · Antiderivative on an interval

A function F is an antiderivative of f on I if F is differentiable on I and $F' = f$ on I. The interval is part of the data. Finding an antiderivative is an inverse calculation; differentiation makes it possible to check it.

Worked example · Correcting a factor

For $f(x) = 3e^{2x}$, $F(x) = \frac{3}{2}e^{2x}$ works because $F'(x) = \frac{3}{2} \times 2e^{2x} = 3e^{2x}$. For $g(x) = 2x/(1+x^2)$ on $\mathbb{R}$, the antiderivative $G(x) = \ln(1+x^2)$ uses a positive argument. All antiderivatives are obtained by adding a constant.

A1 · Immediate application Determine an antiderivative of $x^3 - 2x + 1$, of e^{-3x} and of $1/(2x+1)$ on a suitable interval. Check by differentiation.

7.2 A constant, a vertical translation

First use: S52.

Proof. If $F' = G' = f$ on an interval I, then $(F-G)' = 0$ on I. A function whose derivative is zero on an interval is constant; therefore $F - G = C$. On a domain made of several disjoint intervals, the constant may differ on each interval. □

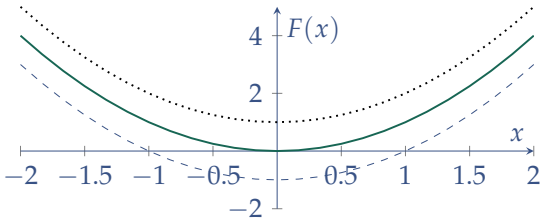


Figure 7.1. Three antiderivatives of $2x$: at the same x-values, the tangents have the same slopes.

Worked example · One value fixes the translation

The antiderivatives of $2x$ are $x^2 + C$. Imposing F(0)=-1 selects $F(x) = x^2 - 1$. The graph is translated one unit downwards without changing the slopes.

A2 · Partial guidance Determine the antiderivative of $2x$ which equals 3 at 1. Then explain why all the antiderivatives of $1/x$ on $\mathbb{R} \setminus \{0\}$ are not described by a single constant common to both half-lines.

7.3 The unknown is a function

First use: S55.

Property · Equation $y' = ay$

For $a \in \mathbb{R}$, the solutions on $\mathbb{R}$ are $y(t) = Ce^{at}$, with $C \in \mathbb{R}$. An initial value $y(0) = y_0$ imposes $C = y_0$.

Proof. A function of this form is indeed a solution. Conversely, if y is a solution, $z(t) = y(t)e^{-at}$ has derivative $(y' - ay)e^{-at} = 0$, so it is constant on the interval. Thus $y(t) = Ce^{at}$. The proof checks the solutions and shows that none is missing. □

Worked example · A condition at another time

For $y' = 3y$ and y(1)=2, $y(t) = Ce^{3t}$, then $Ce^3 = 2$. Hence $C = 2e^{-3}$ and $y(t) = 2e^{3(t-1)}$. The constant equals the prescribed value only if that value is given at 0.

A3 · Immediate application Solve $y' = -2y$ with $y(0) = 5$. Is the number 0 a possible solution of the differential equation alone? Of this initial value problem?

7.4 Subtract a known solution

First use: S55 and S58.

Worked example · Variable right-hand side

For $y' = -y + 2t + 1$, look for a particular solution $p(t) = at + b$ leads to $a = -at - b + 2t + 1$. Identifying the coefficients gives $a = 2$, $b = -1$. If y is a solution, $z = y - p$ satisfies $z' = -z$. With $y(0) = 3$, we get $z(0) = 4$, so $y(t) = 2t - 1 + 4e^{-t}$.

A4 · Partial guidance Solve $y' = 2y - 4$ with $y(0) = 3$ by subtracting a constant solution. Check the formula and initial value.

A5 · Fix a proof A student checked that a function satisfies a differential equation, then wrote "so I have found all the solutions". Why is this conclusion insufficient? Illustrate with $y' = y$.

7.5 Comparing the discrete, the continuous and an approximation

First use: S55 and S58.

Worked example · Cooling

The model $T' = -0{,}1(T - 20)$, with $T(0) = 80$ and t in minutes, gives $T(t) = 20 + 60e^{-0{,}1t}$. With a step h, Euler's method gives $T_{n+1} = T_n - 0{,}1h(T_n - 20)$ at the times $t_n = nh$. For $h = 2$, we get 80, 68, 58.4. These are approximations of the continuous model at times 0, 2, 4; a discrete physical law given independently is not automatically an Euler scheme.

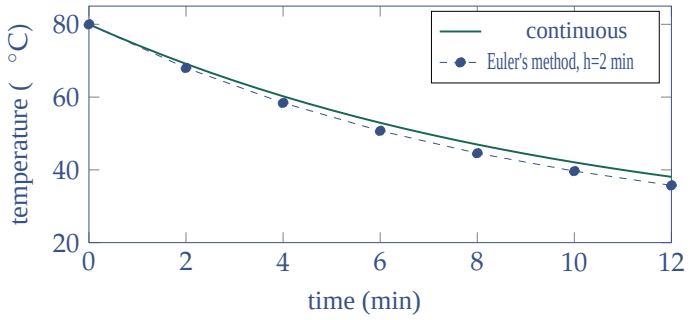


Figure 7.2. Read the gap at t=6 min; the segments join the successive approximations.

A6 · Choice of strategy Determine the first time the continuous model reaches 50 degrees Celsius. Compare with the first value from Euler's method, with h=2 min, which is less than or equal to 50. Explain why the two results differ.

A first step: First solve an exponential equation; for Euler, write the formula for $T_n - 20$.

One minute, spoken aloud. State what y, y' and t denote, then check that the units of both sides of a differential equation are compatible.

Delayed review · S78. Complete the table of antiderivatives with trigonometric functions and check the factors of the compositions.

Investigation

Cooling a drink - 40 min.

A drink initially at 80 °C is placed in a room kept at 20 °C. For $t \geq 0$, let $T(t)$ be its temperature in degrees Celsius, t denoting the time elapsed in minutes. We assume the model $T'(t) = -0{,}1(T(t) - 20)$ and $T(0) = 80$.

1. Solve this initial value problem and give the expression of $T(t)$.
2. Study the direction of variation of T and determine its limit at $+\infty$. Interpret the results.
3. Determine the first time the temperature is 40 °C. Give the exact value, then an approximate value in minutes.
4. Apply two steps of Euler's method with a step of 1 minute. Compare the approximation obtained after two minutes with the exact value $T(2)$.
5. Give a concrete situation in which the assumption of a constant ambient temperature would no longer be suitable.